



Course Outline

1. Course Identity

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	MAT2001
Course title (English)	Honours Ordinary Differential Equations
Course title (Chinese)	常微分方程荣誉课程
Units	3
Language of Instruction	English
Description (English)	This course teaches fundamental principles and methods of ordinary differential equations with rigorous theory and sophisticated techniques. It covers the topics of first order differential equations, second order linear equations, two point boundary-value problems, systems of linear equations, and qualitative theory of nonlinear differential equations, as well as their applications.
Description (Chinese)	本课程教授常微分方程的基本原理和方法，包括严格的理论和精巧的技术。课程内容包括一阶方程，二阶线性方程，线性方程组，与非线性方程的定性理论，以及它们的应用。

2. Prerequisites / Co-requisites

A. Prerequisites

MAT1001 and MAT1002, or MAT1011 and MAT1012
MAT2040 Linear Algebra

B. Co-requisites

3. Learning Outcomes

Upon successful completion of the courses, the students will be able to understand basic theory of ODE (ordinary differential equations), master basic techniques in ODE, and handle nonlinear ODE by using qualitative theory of ODE.



Knowledge

- Basic theory of first order linear and nonlinear ordinary differential equations.
- General theory of higher order linear ordinary differential equations.
- Basic theory of boundary value problems.
- Basic theory of systems of ordinary differential equations.
- Qualitative theory of nonlinear ordinary differential equations and its applications

Skill

- Ability of carrying out mathematical modelling, solving first order linear and nonlinear equations.
- Ability of solving second order linear equations.
- Ability of solving systems of ordinary differential equations.
- Ability of solving two point boundary value problems.
- Ability of analyzing stability of nonlinear ordinary differential equations, finding periodic solutions.

4. Course syllabus

Introduction: mathematical modelling, solutions of differential equations

First Order Differential Equations: linear equations, separable equations, exact equations, integrating factors, explicitly solvable equations, population models, pursue problems

Existence and Uniqueness of solutions for initial value problems

Second Order Linear Equations: solutions for linear second order equations with constant coefficients, characteristic equation, Wronskian, method of undetermined coefficients, variations of parameters, mechanical vibrations, damping effects, resonance, and applications, series solutions, two point boundary-value problems

Higher Order Linear Equations: general theory, homogeneous equations

Systems of ODEs: first order systems, solving systems using linear algebra, eigenvalues, fundamental solutions, complex eigenvalues, repeated eigenvalues

Qualitative Theory: phase portraits for linear systems, concept of stability, method of linearization, Liapunov's function, applications to population models, periodic solutions, limit cycles, Poincare-Bendixon Theorem, Lotka-Volterra Competition Systems, Prey-Predator System

5. Assessment Scheme

Component/ method	% weight
Homework	25
Mid-Term Exam	25



Final Exam	50
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6. Grade descriptor

General

Grade	Description
A	Demonstrates the ability to synthesize and apply the principles or subject matter learnt in the course, to novel situations and/or in novel ways, in a manner that would surpass the normal expectation at this level, and typical of standards that may be common at higher levels of study or research. Has the ability to express the synthesis of ideas or application in a clear and cogent manner.
B	Demonstrates the ability to state and partially apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is usually logically persuasive. Has the ability to express the knowledge or application in a satisfactory and unambiguous way.
C	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is not incorrect but is somewhat fragmented. Has the ability to express the separate pieces of knowledge in an unambiguous way.
D	Demonstrates the ability to state and sometimes apply the principles or subject matter learnt in the course to some simple and familiar situations in a manner that is broadly correct in its essentials Has the ability to state the knowledge or application in simple terms.
F	Unsatisfactory performance on a number of learning outcomes, OR failure to meet specified assessment requirements.

7. Feedback for evaluation

Collect student feedback from CTE by the end of the course
Collect opinions from peers by internal and external course reviews

8. Reading

Required Reading



1. Differential Equations and their Applications, 4, Germany

9. Course components

Activity	Hours/week
Lecture	3
Tutorial	1

10. Indicative teaching plan

Week	Content/ topic/ activity
00000 00001	First order linear differential equations. Separable equations. Population models.
00000 00002	Applications – Newton's cooling law, pursue problems.
00000 00003	Existence and uniqueness
00000 00004	Second order linear equations with constant coefficients.
00000 00005	Applications – mechanical vibrations.
00000 00006	Series solutions
00000 00007	Fundamental matrix. Nonhomogeneous equation. Variation of parameters.
00000 00008	Stability of linear systems.
00000 00009	Stability of equilibrium solutions for nonlinear systems.
00000 00010	The phase plane. Qualitative properties of orbits.
00000 00011	Phase portraits of linear systems.
00000 00012	Long time behavior of solutions
00000 00013	Poincare-Bendixson Theorem and applications
00000 00014	Prey-predator system. Lotka-Volterra competition system. Epidemic models.

11. Any other information



Prohibit the use of AI tools: AI tools (particularly GPT and generative AI tools) are not allowed in submitted reports. Any student who violates this rule will be considered to be engaging in academic dishonesty. The instructor may specify which AI tools are prohibited in this course

12. Version date

Version number	4
As of (date)	2024/08/12